Delta = c_s(N) / c_s^KR - 1 (E1) against u = N^(-1/2) per density, with the finite-size fits of sec. 4.8.7.1
Verdict K on Phi: KR-INCONSISTENT, X^2 = 32.69 (12 dof), p = 0.0011, max |z| = 3.58

N = 100 | N = 400 | N = 900 | N = 1600 | F2 = a + b u + d u^2 (primary), a +- sigma_a at u = 0 | F1 = a + b u (reported) | KR (Delta = 0)

eta = 0.05 (Phi)
F2: a = +0.049 +- 0.080 %, z = +0.62, p = 0.580
F1: a = -0.110 +- 0.030 %, p = 0.086

eta = 0.1 (Phi)
F2: a = -0.026 +- 0.068 %, z = -0.38, p = 0.092
F1: a = -0.043 +- 0.026 %, p = 0.233

eta = 0.15 (Phi)
F2: a = -0.090 +- 0.065 %, z = -1.38, p = 0.049
F1: a = +0.003 +- 0.023 %, p = 0.046

eta = 0.2 (Phi)
F2: a = -0.014 +- 0.067 %, z = -0.21, p = 0.645
F1: a = +0.090 +- 0.021 %, p = 0.237

eta = 0.25 (Phi)
F2: a = +0.084 +- 0.059 %, z = +1.42, p = 0.437
F1: a = +0.164 +- 0.021 %, p = 0.262

eta = 0.3 (Phi)
F2: a = +0.004 +- 0.058 %, z = +0.06, p = 0.563
F1: a = +0.183 +- 0.022 %, p = 0.003

eta = 0.35 (Phi)
F2: a = +0.023 +- 0.078 %, z = +0.29, p = 0.209
F1: a = +0.282 +- 0.024 %, p = 0.001

eta = 0.3927 (Phi)
F2: a = +0.222 +- 0.086 %, z = +2.58, p = 0.126
F1: a = +0.441 +- 0.024 %, p = 0.009

eta = 0.45 (Phi)
F2: a = +0.151 +- 0.099 %, z = +1.52, p = 0.699
F1: a = +0.527 +- 0.038 %, p = 0.000

eta = 0.5 (Phi)
F2: a = +0.244 +- 0.097 %, z = +2.50, p = 0.004
F1: a = +0.336 +- 0.034 %, p = 0.011

eta = 0.55 (Phi)
F2: a = -0.003 +- 0.099 %, z = -0.03, p = 0.024
F1: a = +0.327 +- 0.041 %, p = 0.000

eta = 0.6 (Phi)
F2: a = -0.537 +- 0.150 %, z = -3.58, p = 0.012
F1: a = +0.353 +- 0.066 %, p = 0.000

eta = 0.63 (approach cell: no verdict)
F2: a = -0.423 +- 0.241 %, z = -1.76, p = 0.107
F1: a = -0.527 +- 0.103 %, p = 0.243

eta = 0.66 (approach cell: no verdict)
F2: a = -5.011 +- 0.816 %, z = -6.14, p = 0.107
F1: a = -6.700 +- 0.196 %, p = 0.028

eta = 0.68 (approach cell: no verdict)
F2: a = -17.164 +- 1.741 %, z = -9.86, p = 0.006
F1: a = -11.728 +- 0.440 %, p = 0.000

eta = 0.69 (approach cell: no verdict)
F2: a = -21.355 +- 1.234 %, z = -17.31, p = 0.000
F1: a = -14.479 +- 0.379 %, p = 0.000

Delta [%]

u = N^(-1/2) (0.1, 0.05, 0.033, 0.025 = N 100, 400, 900, 1600)

data: 261015_s1_fig1_cs_vs_eta_E1_N100-1600.csv (the points), 261015_s1_fig2_Delta_vs_Nm12_per_eta_F1_F2_fits.csv (the fits)
script: cluster/gen3_stage1_261012/s1_figures_261015.py (s1_verdict_261012.analyse on experiments_gen3_stage1_koa_261012, all six groups, the full report's numbers)
N = 100, 400, 900, 1600 disks, N/2 per side; disk radius 0.5; divider thickness 0.05; H = 10 sqrt(N/100) = 10, 20, 30, 40; L0 = N pi / (8 H eta) per side (exact box); gen3 engine, one build (f72d764); 25 seeds per (cell, mass), 10 in block S